

Muhammad Huzaifa Khan

Assignment # 9-10
JAVASCRIPT

1. Write a program to take “city” name as input from user. If user enters “Karachi”, welcome the user like this:
“Welcome to city of lights”

```
var city = prompt("Enter your city name:");
```

```
if (city.toLowerCase() === "karachi") {  
  alert("Welcome to city of lights");  
}
```

OUTPUT:

Welcome to city of lights

2. Write a program to take “gender” as input from user. If the user is male, give the message: Good Morning Sir. If the user is female, give the message: Good Morning Ma’am.

```
var gender = prompt("Enter your gender (Male/Female):");
```

```
if (gender.toLowerCase() === "male") {  
  alert("Good Morning Sir.");  
}  
else if (gender.toLowerCase() === "female") {  
  alert("Good Morning Ma'am.");  
}
```

OUTPUT:

Good Morning Sir
Good Morning Ma'am

3. Write a program to take input color of road traffic signal from the user & show the message according to this table:

Signal color	Message
Red	Must Stop
Yellow	Ready to move
Green	Move now

Code:

```
var color = prompt("Enter traffic signal color:");

if (color.toLowerCase() === "red") {
  alert("Must Stop");
}
else if (color.toLowerCase() === "yellow") {
  alert("Ready to move");
}
else if (color.toLowerCase() === "green") {
  alert("Move now");
}
else {
  alert("Invalid Color");
}
```

OUTPUT:

Red → Must Stop
Yellow → Ready to move
Green → Move now

4. Write a program to take input remaining fuel in car (in litres) from user. If the current fuel is less than 0.25litres, show the message “Please refill the fuel in your car”

```
var fuel = parseFloat(prompt("Enter remaining fuel in litres:"));
```

```
if (fuel < 0.25) {  
  alert("Please refill the fuel in your car");  
}
```

OUTPUT:

Please refill the fuel in your car

5. Run this script, & check whether alert message would be displayed or not. Record the outputs.

```
a. var a = 4;  
if (++a === 5){  
  alert("given condition for variable a is true");  
}
```

```
b. var b = 82;
if (b++ === 83){
  alert("given condition for variable b is true");
}
```

```
c. var c = 12;
if (c++ === 13){
  alert("condition 1 is true");
}
if (c === 13){
  alert("condition 2 is true");
}
if (++c < 14){
  alert("condition 3 is true");
}
if(c === 14){
  alert("condition 4 is true");
}
```

```
d. var materialCost = 20000;
var laborCost = 2000;
var totalCost = materialCost + laborCost;
if (totalCost === laborCost + materialCost){
  alert("The cost equals");
}
```

```
e. if (true){
  alert("True");
}
if (false){
  alert("False");
}
```

```
f. if("car" < "cat"){
  alert("car is smaller than cat");
}
```

OUTPUT:

```
a: alert shown
b: no alert
c: condition 2 & 4 true
d: The cost equals
e: True
f: car is smaller than cat
```

6. Write a program to take input the marks obtained in three subjects & total marks. Compute & show the resulting percentage on your page. Take percentage & compute grade as per following table:

Percentage %	Grade	Remarks
Greater than or equal to 80	A-one	Excellent
Greater than or equal to 70	A	Good
Greater than or equal to 60	B	You need to improve
Less than 60	Fail	Sorry

Show the total marks, marks obtained, percentage, grade & remarks like:

Marks Sheet

Total marks : 300

Marks obtained : 219

Percentage : 73%

Grade : B

Remarks : You need to improve

```
var totalMarks = parseInt(prompt("Enter Total Marks:"));

var sub1 = parseInt(prompt("Enter Marks of Subject 1:"));
var sub2 = parseInt(prompt("Enter Marks of Subject 2:"));
var sub3 = parseInt(prompt("Enter Marks of Subject 3:"));

var obtained = sub1 + sub2 + sub3;

var percentage = (obtained / totalMarks) * 100;

var grade;
var remarks;

if (percentage >= 80) {
  grade = "A-one";
  remarks = "Excellent";
}
else if (percentage >= 70) {
  grade = "A";
  remarks = "Good";
}
else if (percentage >= 60) {
  grade = "B";
  remarks = "You need to improve";
}
else {
  grade = "Fail";
  remarks = "Sorry";
}

document.write("<h2>Marks Sheet</h2>");
document.write("Total Marks: " + totalMarks + "<br>");
```

```
document.write("Marks Obtained: " + obtained + "<br>");  
document.write("Percentage: " + percentage.toFixed(2) + "%<br>");  
document.write("Grade: " + grade + "<br>");  
document.write("Remarks: " + remarks);
```

OUTPUT:

Sample:
Total:300
Obtained:240
80%
Grade:A
Remarks:Excellent

7. Guess game:

Store a secret number (ranging from 1 to 10) in a variable.

Prompt user to guess the secret number.

a. If user guesses the same number, show “Bingo! Correct answer”.

b. If the guessed number +1 is the secret number, show “Close enough to the correct answer”.

```
var secret = 7;
```

```
var guess = parseInt(prompt("Guess the secret number (1-10):"));
```

```
if (guess === secret) {
  alert("Bingo! Correct answer");
}
else if (guess + 1 === secret) {
  alert("Close enough to the correct answer");
}
else {
  alert("Try Again!");
}
```

OUTPUT:

Bingo! Correct answer
or Close enough to the correct answer

8. Write a program to check whether the given number is divisible by 3. Show the message to the user if the number is divisible by 3.

```
var num = parseInt(prompt("Enter a number:"));
```

```
if (num % 3 === 0) {
  alert("The number is divisible by 3");
}
else {
```

```
alert("The number is not divisible by 3");
}
```

OUTPUT:

9 is divisible by 3

9. Write a program that checks whether the given input is an even number or an odd number.

```
var number = parseInt(prompt("Enter a number:"));

if (number % 2 === 0) {
  alert("Even Number");
}
else {
  alert("Odd Number");
}
```

OUTPUT:

8 is Even

10. Write a program that takes temperature as input and shows a message based on following criteria
- T > 40 then "It is too hot outside."
 - T > 30 then "The Weather today is Normal."
 - T > 20 then "Today's Weather is cool."
 - T > 10 then "OMG! Today's weather is so Cool."

```
var temp = parseInt(prompt("Enter Temperature:"));
```

```
if (temp > 40) {  
  alert("It is too hot outside.");  
}  
else if (temp > 30) {  
  alert("The Weather today is Normal.");  
}  
else if (temp > 20) {  
  alert("Today's Weather is cool.");  
}  
else if (temp > 10) {  
  alert("OMG! Today's weather is so Cool.");  
}  
else {  
  alert("It is very cold today.");  
}
```

OUTPUT:

35°C → The Weather today is Normal.

11. Write a program to create a calculator for +, -, *, / & % using if statements. Take the following input:
- First number
 - Second number
 - Operation (+, -, *, /, %)
- Compute & show the calculated result to user.

```
if (operator === "+") {  
  alert("Result = " + (num1 + num2));  
}  
else if (operator === "-") {  
  alert("Result = " + (num1 - num2));  
}  
else if (operator === "*") {  
  alert("Result = " + (num1 * num2));  
}  
else if (operator === "/") {  
  alert("Result = " + (num1 / num2));  
}  
else if (operator === "%") {  
  alert("Result = " + (num1 % num2));  
}  
else {  
  alert("Invalid Operator");  
}
```

OUTPUT:

10 + 5 = 15